Mathematics GCE O Level 2021 - Paper 2

Duration: 2h30 | Cameroon GCE Board

This paper is arranged in two sections, A and B. Answer ALL questions in Sections A and B. Section A: answer ALL the questions (15 questions); the mark allocation for each question is indicated. Section B: answer ALL four questions; all questions in Section B carry equal marks (15 marks each). You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed. Duration: two and a half hours.

Exercise 1 (5 marks)

During an auction sale, a shop reduced the prices of its articles by 10%. Calculate the selling price of an article originally priced at 43000 FCFA. *[5 marks]*

Exercise 2 (4 marks)

Solve for x in the equation $\frac{5}{7x-1} = \frac{4}{9}$. *[4 marks]*

Exercise 3 (5 marks)

Given an equation of a straight line $\frac{x}{2} + \frac{y}{4} = 1$. Determine:

a. The gradient of the line.

b. The coordinates of the point where the line meets the y-axis. *[5 marks for the whole question]*

Exercise 4 (5 marks)

Given the function $f(x) = x^2 - 4x + 3$.

Find:

a. The factors of $f(x)$.

b. Hence, the roots of $f(x)$. *[5 marks for the whole question]*

Exercise 5 (5 marks)

Draw the truth table for the compound statement $\sim (p \wedge q)$. *[5 marks]*

Exercise 6 (6 marks)

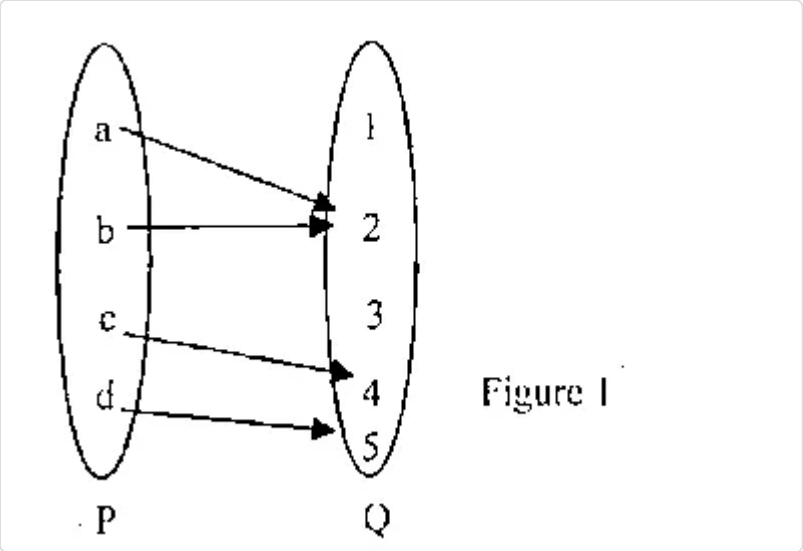


Figure 1

Figure 1 shows a relation $P \to Q$.

a. State whether this is a one - one, one - many or many - one relation.

b. Write down the Cartesian product $P \times Q$ of the sets P and Q, for which the relation is true.

c. State the range of the relation. *[6 marks for the whole question]*

Exercise 7 (6 marks)

Given the matrix $M = \begin{pmatrix} 1 & 2 \\ x-1 & x \end{pmatrix}$.

Find:

a. The transpose M^T.

b. The value of x for which $M = M^T$.

c. Using this value of x, find M^2. *[6 marks for the whole question]*

Exercise 8 (6 marks)

Given that y is directly proportional to x and that $y = 24$ when $x = 3$.

Find:

a. An equation in terms of x and y.

b. The value of y when $x = 7$. *[6 marks for the whole question]*

Exercise 9 (6 marks)

An interior angle of a regular polygon is twice its exterior angle.

Calculate:

a. The value of an exterior angle.

b. The number of sides of the polygon.

c. Name the polygon. *[6 marks for the whole question]*

Exercise 10 (5 marks)

Solve the inequality $-2(5 - x) \leq 4 - 5x$ and represent your solution on a real number line. *[5 marks]*

Exercise 11 (6 marks)

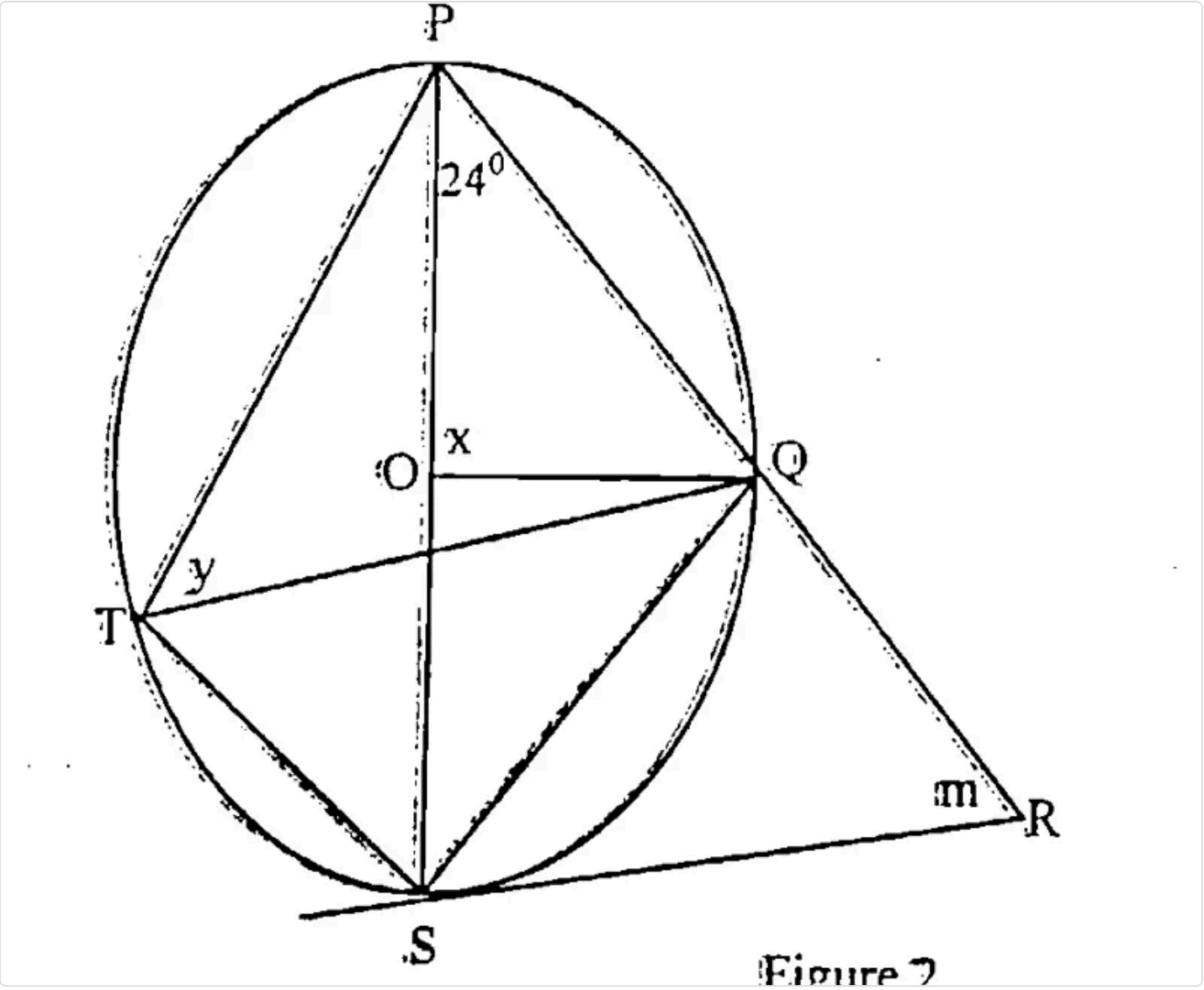


Figure 2

Figure 2 is a circle with centre O and SR is a tangent. Given that angle $OPQ = 24°$.

Find the value of:

a. Angle x.

b. Angle y.

c. Angle m. *[6 marks for the whole question]*

Exercise 12 (6 marks)

Given the vectors $\overrightarrow{OP} = (a+2)\mathbf{i} + \mathbf{j}$ and $\overrightarrow{OQ} = 2a\mathbf{i} + 5\mathbf{j}$, where a is a constant.

a. Find $\overrightarrow{PQ}$ in terms of a.

b. Given that $\overrightarrow{PQ} = 3\mathbf{i} + 4\mathbf{j}$, find the value of a.

c. Find $\left|\overrightarrow{PQ}\right|$. *[6 marks for the whole question]*

Exercise 13 (6 marks)

a. Draw a network with 4 regions and 4 vertices.

b. From the network, find the number of arcs. *[6 marks for the whole question]*

Exercise 14 (7 marks)

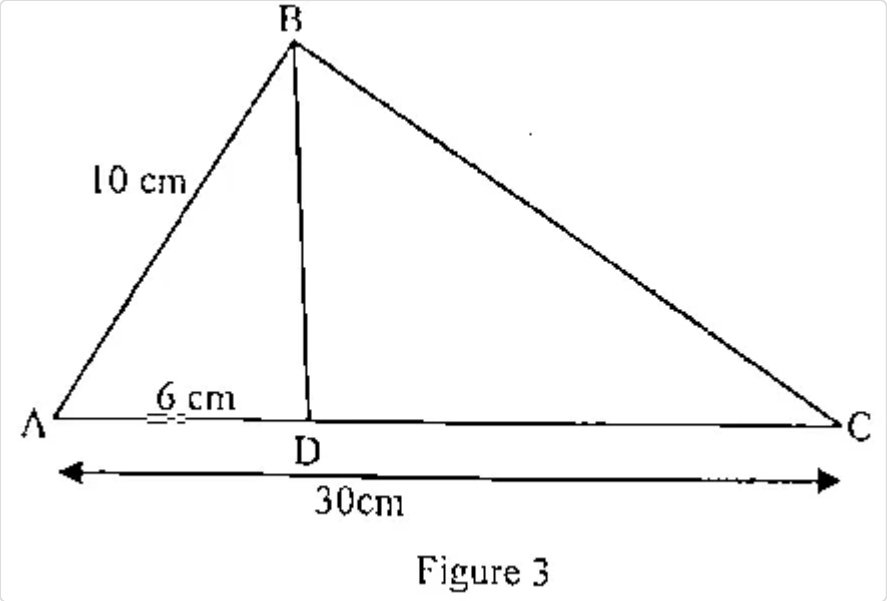


Figure 3

In figure 3, ABC is a triangle in which BD is a perpendicular to AC. Given that $AC = 30$ cm, $AB = 10$ cm and $AD = 6$ cm.

Find:

a. BD.

b. DC.

c. The angle DCB. *[7 marks for the whole question]*

Exercise 15 (5 marks)

Given that the events A and B are mutually exclusive.

a. State the condition for which A and B are mutually exclusive.

b. Hence, given that $P(A) = \frac{1}{6}$ and $P(B) = \frac{3}{4}$, find $P(A \cup B)$.

c. $P(A^c)$ (with $P(A) = \frac{1}{6}$ as above). *[5 marks for the whole question]*

Miriam sold Rice, Beans, Corn and Garri in buckets on a particular day. She made a profit per bucket for each item sold as follows:

Rice: 600 FCFA, Beans: 500 FCFA, Corn: 300 FCFA and the profit for Garri was $\frac{2}{3}$ of the profit for Rice.

Find:

a. The profit per bucket of Garri on that day.

b. The ratio of the profit per bucket of Rice and Garri sold.

c. The total profit she made on that day from selling 4 buckets of Rice, 2 buckets of Beans, 5 buckets of Corn and 4 buckets of Garri.

d. She used $\frac{2}{5}$ of her profit to buy fish. Calculate the amount spent to buy fish.

e. She used $\frac{1}{13}$ of the remainder of her profit for her taxi fare and saved the rest. Find her taxi fare.

f. Find the amount saved.

(i) Given the function $f(x) = 5 + 4x - x^2$.

(ii) A function f is defined on $\mathbb{R}$, the set of real numbers, by $f(x) = x^3 + 2x^2 + ax + b$. When $f(x)$ is divided by $(x + 1)$, the remainder is 0.

a. (i)(a) Construct a table of values of y against x for $-2 \leq x \leq 6$.

b. (i)(b) Taking 2 cm to represent 1 unit on the x-axis and 1 cm to represent 1 unit on the y-axis, draw the graph of $y = f(x)$.

c. (i)(c) From your graph, determine the value(s) of x for which $x^2 - 4x - 5 = 0$.

d. (i)(d) Find the gradient of the curve at the point $(4, 5)$, giving your answer to 1 decimal place.

e. (ii)(a) Given that $(x - 2)$ is a factor of $f(x)$, find the values of a and b.

f. (ii)(b) Factorize $f(x)$ completely.

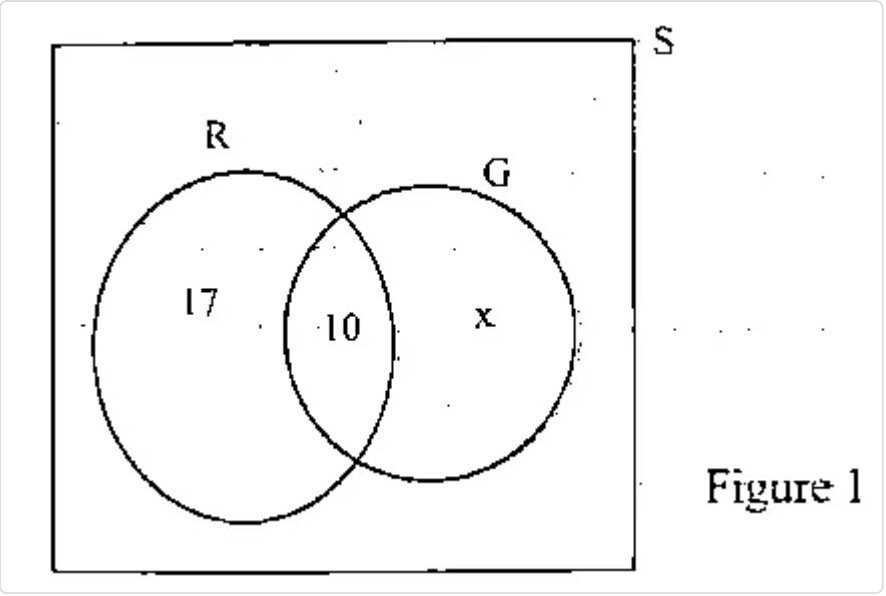


Figure 1

(i) Figure 1 is a Venn diagram showing 50 students in a boarding school who ate Rice (R) and Garri (G) for lunch on a particular day. All the students ate either Rice or Garri or both.

(ii) Using ruler, pencil and a pair of compasses only:

a. (i)(a) Find the number of students who ate Rice.

b. (i)(b) Find the value of x.

c. (i)(c) Describe in ordinary English the region with the value 17.

d. (i)(d) In set notation, give the number of students who ate both Rice and Garri.

e. (i)(e) In set notation, describe the region with no student.

f. (ii)(a) Construct a triangle PQR in which $PQ = 10$ cm, $QR = 9$ cm and $PR = 7$ cm.

g. (ii)(b) Construct the bisectors of angles PQR and PRQ.

h. (ii)(c) Mark the point M, where the bisectors meet.

i. (ii)(d) Construct a circle with centre M such that PQ, QR and PR are tangents to the circle.

j. (ii)(e) Measure and write down the radius.

The table below shows the results of candidates who wrote the ordinary level examination in a certain school.

No of subjects passed	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
No of candidates passed	3	4	6	10	21	20	10	6	8	7	3	2

a. State the mode.

b. Find the total number of candidates that sat for the examination.

c. Find the median.

d. Find the mean number of subjects passed, giving your answer to the nearest whole number.

e. Given that a candidate who passed in four or more subjects has passed the examination, find the number of candidates who passed the examination.

f. Find the probability that a candidate chosen at random passed the examination.

g. Calculate the percentage pass to one decimal place.

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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